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November 10, 2025, 2:08 PM

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CAS Lexicon Search:

Rosenmund reduction - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (12)	 References	View Results
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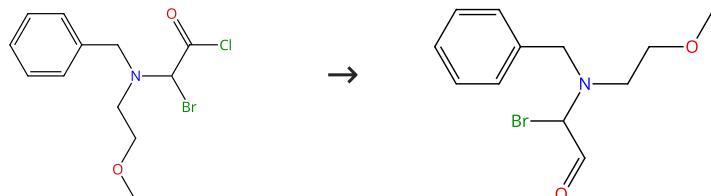
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Reactions (17)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 97%



31-516-CAS-16295009

Steps: 1 Yield: 97%

Method for preparation of aprepitant bulk drug intermediate with optical purity

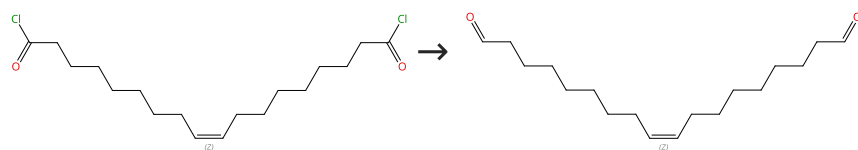
 Assignee: Alpha Pharmaceutical Co., Ltd., Jiangsu Province
 China, CN104496930 A 2015-04-08

[PatentPak](#) available

 1.1 **Reagents:** Hydrogen
Catalysts: Lead, Barium sulfate
Solvents: Toluene; 4 h, rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Double bond geometry shown

31-516-CAS-16899627

Steps: 1 Yield: 96%

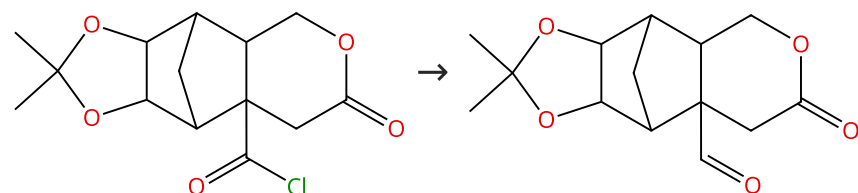
Palladium on barium sulfate

 By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis
 (2001), 1-4.

 1.1 **Reagents:** 2,6-Lutidine, Hydrogen
Catalysts: Palladium
Solvents: Tetrahydrofuran; 1 atm, 25 °C

Scheme 3 (1 Reaction)

Steps: 1 Yield: 95%



31-516-CAS-16899621

Steps: 1 Yield: 95%

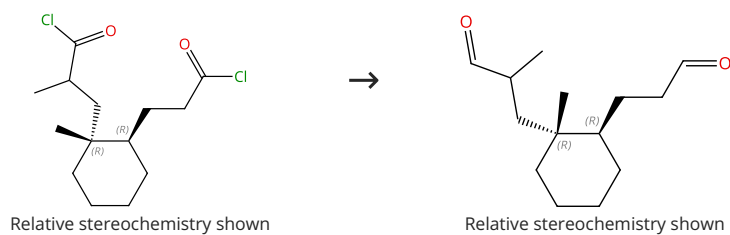
Palladium on barium sulfate

 By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis
 (2001), 1-4.

 1.1 **Reagents:** Hydrogen
Catalysts: Palladium
Solvents: Toluene; reflux

Scheme 4 (1 Reaction)

Steps: 1 Yield: 94%



31-516-CAS-16899624

Steps: 1 Yield: 94%

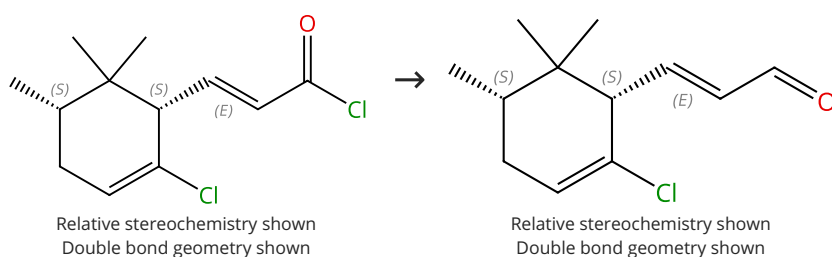
Palladium on barium sulfate

1.1 **Reagents:** Hydrogen
Catalysts: Palladium
Solvents: Benzene; 1 atm, 30 - 35 °C

By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 93%



31-516-CAS-16899629

Steps: 1 Yield: 93%

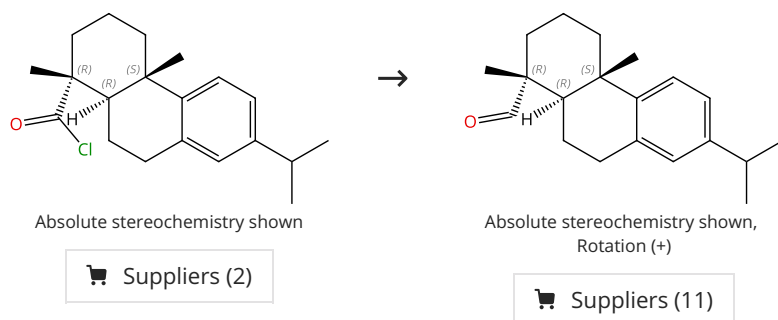
Palladium on barium sulfate

1.1 **Reagents:** Hydrogen
Catalysts: Palladium
Solvents: Toluene; reflux

By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 93%



31-516-CAS-16899628

Steps: 1 Yield: 93%

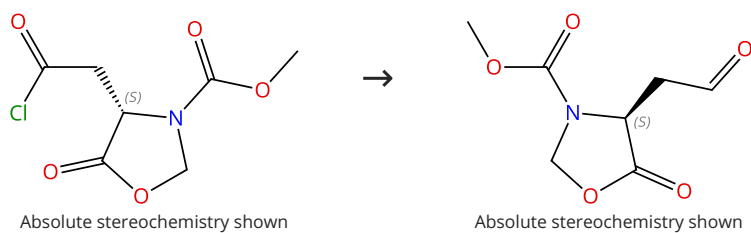
Palladium on barium sulfate

1.1 **Reagents:** 2,6-Lutidine, Hydrogen
Catalysts: Palladium
Solvents: Tetrahydrofuran; 1 atm, 0 °C

By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (3)

31-516-CAS-16899632

Steps: 1 Yield: 93%

Palladium on barium sulfate

1.1 Reagents: Hydrogen
Catalysts: Palladium
Solvents: Toluene; reflux

By: Siegel, Samuel; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis
(2001), 1-4.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (45)

Suppliers (68)

31-614-CAS-24730626

Steps: 1 Yield: 90%

Preparation of transfluthrin intermediate

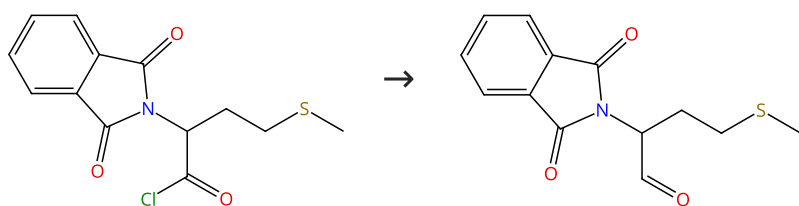
1.1 Reagents: Thiourea, Hydrogen
Catalysts: Calcium carbonate, Palladium; 6 h, 60 °C

Assignee: Zhejiang Zhongxin Fluorine Materials Co., Ltd.
China, CN113292399 A 2021-08-24

[PatentPak](#) available

Scheme 9 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (2)

31-516-CAS-16899623

Steps: 1 Yield: 90%

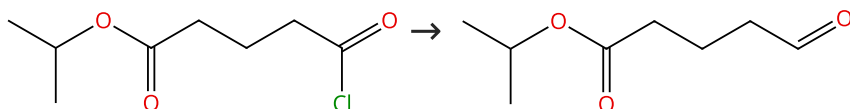
Palladium on barium sulfate

1.1 Reagents: Hydrogen
Catalysts: Palladium
Solvents: Benzene; 38 - 40 °C

By: Siegel, Samuel; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis
(2001), 1-4.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (4)

Suppliers (5)

31-516-CAS-5160004

Steps: 1 Yield: 89%

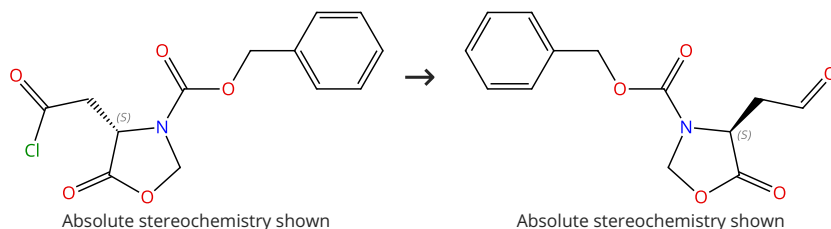
2,6-Lutidine

1.1 **Reagents:** 2,6-Lutidine, Hydrogen
Catalysts: Palladium
Solvents: Tetrahydrofuran; 1 - 2 h, rt

By: Sampson, Paul; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis
 (2007), 1-6.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 87%



Suppliers (3)

Suppliers (4)

31-516-CAS-16899630

Steps: 1 Yield: 87%

Palladium on barium sulfate

1.1 **Reagents:** Hydrogen
Catalysts: Palladium
Solvents: Toluene; reflux

By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis
 (2001), 1-4.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (68)

Suppliers (85)

31-516-CAS-16899625

Steps: 1 Yield: 83%

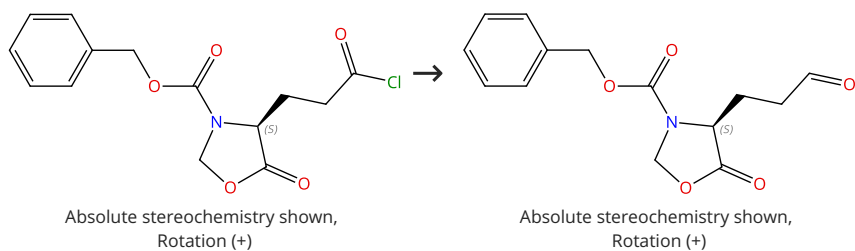
Palladium on barium sulfate

1.1 **Reagents:** Quinoline, Sodium acetate, Hydrogen, Sulfur
Catalysts: Palladium
Solvents: Toluene; 4 atm, 35 - 40 °C

By: Siegel, Samuel; et al
 e-EROS Encyclopedia of Reagents for Organic Synthesis
 (2001), 1-4.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (3)

Suppliers (3)

31-516-CAS-16899631

Steps: 1 Yield: 82%

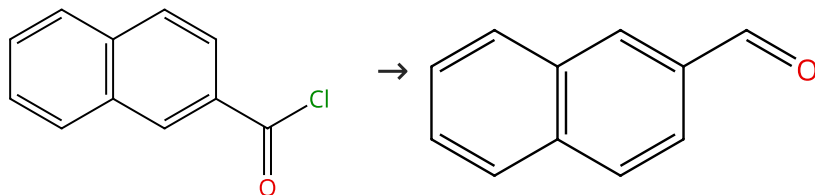
Palladium on barium sulfate

1.1 Reagents: Hydrogen
Catalysts: Palladium
Solvents: Toluene; reflux

By: Siegel, Samuel; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.

Scheme 14 (2 Reactions)

Steps: 1 Yield: 81%



Suppliers (63)

Suppliers (86)

31-516-CAS-16075069

Steps: 1 Yield: 81%

Quinoline

1.1 Reagents: Hydrogen
Catalysts: Quinoline, Palladium, Sulfur, Barium sulfate

By: Sherman, Angela R.; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2015), 1-4.

31-516-CAS-16899618

Steps: 1 Yield: 81%

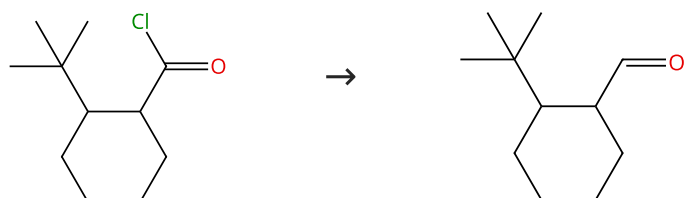
Palladium on barium sulfate

1.1 Reagents: Quinoline, Hydrogen, Sulfur
Catalysts: Palladium
Solvents: Xylene; reflux

By: Siegel, Samuel; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.

Scheme 15 (1 Reaction)

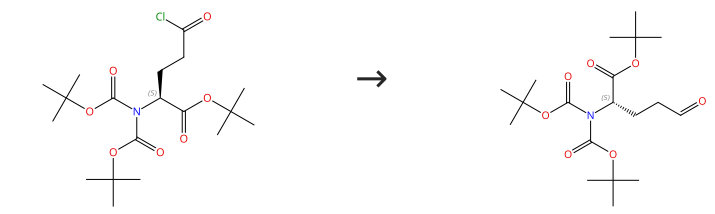
Steps: 1 Yield: 78%



Suppliers (5)

Suppliers (2)

31-516-CAS-16899626	Steps: 1 Yield: 78%	Palladium on barium sulfate
1.1 Reagents: Hydrogen, Diisopropylethylamine Catalysts: Palladium Solvents: Acetone; 1 atm, 25 °C	By: Siegel, Samuel; et al e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.	

Scheme 16 (1 Reaction)			Steps: 1 Yield: 78%
 A chemical reaction scheme showing the conversion of a chiral intermediate to a product. The reactant is a complex molecule with a central carbon atom bonded to a chlorine atom (green), a tert-butyl ester group, and a chiral auxiliary. The product is a similar molecule but with a different stereochemistry at the chiral center, indicated by a wavy bond. The reaction is labeled 'Scheme 16 (1 Reaction)'.		Absolute stereochemistry shown Absolute stereochemistry shown, Rotation (-)  Suppliers (16)	

31-516-CAS-16899634	Steps: 1 Yield: 78%	Palladium on barium sulfate
1.1 Reagents: 2,6-Lutidine, Hydrogen Catalysts: Palladium Solvents: Tetrahydrofuran; 1 atm, 10 - 15 °C	By: Siegel, Samuel; et al e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-4.	